

## Public Economics Practicum 3

### Discussion Questions Rosen:

|            | 8th edition | 9th & 10th editions |
|------------|-------------|---------------------|
| Chapter 9  | 5, 6, 7     | 6, 7, 8             |
| Chapter 10 | 4, 5        | 5, 6                |
| Chapter 11 | 8           | 8                   |

### Additional Exercises:

- 3.1 Suppose a group of individuals differs in ability  $a$ . Ability is private information: individuals know their own ability, but companies that might employ these individuals only know that ability is uniformly distributed between 0 and 1. When self-employed, an individual with ability  $a_i$  earns  $5000 \cdot a_i$ . When employed by a company, the individual creates revenue equal to  $5000 \cdot a_i + 1000$ . Assume that companies compete for employees.
- (a) When all potential employees would be working for a company, what is the maximum wage the company would be willing to pay?
  - (b) If all companies offer the wage derived under (a), would all employees choose to work for a company? If not, which employees would choose to be self-employed?
  - (c) In equilibrium, what wage is offered by companies, and which individuals work for a company?
  - (d) Is this outcome efficient? Is there a way to improve upon this outcome (in terms of efficiency)?
- 3.2 Consider the market for second-hand cars. Suppose that second-hand cars are either good or bad, and that the fraction of bad cars is  $\frac{1}{2}$ . A bad car will surely break down within 6 months, a good car never breaks down within 6 months. Sellers know the quality of their cars. A seller's valuation of a good car is 3000 euro; sellers' valuation of a bad car is 1000 euro. Buyers do not know the quality of a given car, but do know that cars can be either good or bad (and, hence, that a seller's valuation is either 1000 or 3000 euro). Suppose first that buyers' valuation of a bad car is 3000 euro (despite breaking down) and of a good car 5000 euro.
- (a) Which cars will be sold, and at which price?

Now suppose that buyers' valuation of a bad car is 1600 euro and of a good car 4000 euro.

- (b) Is it efficient for both car types to be traded?
- (c) Which cars will be sold, and at which price?

Now suppose that there is a third party: the inspectors. By inspecting a car, which costs 336 euro, they can (credibly) determine whether the quality is good or bad.

- (d) If buyers always have the car inspected before they buy it, how does this affect your answer to (c)? Does this make the market fully efficient?
- (e) It is actually not necessary to have all cars inspected. Show that if sellers have to bear the cost of inspecting, sellers of bad cars will honestly declare that their car is bad, such that only good cars are inspected.
- (f)\* It is possible to improve efficiency (i.e. decrease inspection cost) even further by randomly inspecting only a fraction of the cars offered by sellers who claim that their car is good. What is the minimum fraction of cars that have to be inspected when the seller claims that the car is good to make sure that sellers of bad cars optimally declare that their car is bad?
- (g) How would you react to a non-economist arguing: "It is better to buy a car from marktplaats or e-bay than from an official dealer, as thereby you do not have to pay for the profits of the dealer. The benefits are obvious if you compare the prices."

3.3 Consider a small village somewhere in East-Asia, where the only source of income is bamboo, which can be sold after it has been cut. The utility of a bamboo-cutter is given by

$$U = 8\sqrt{Y}$$

where  $Y$  is income. Bamboo-cutting is dangerous, due to the presence of poisonous snakes. However, bamboo-cutters can take precautions, which reduce the probability of being bitten. Without precautions, this probability is  $\frac{1}{2}$ . With precautions, the probability of being bitten is lowered to  $\frac{1}{4}$ . Unfortunately, the precautions are unpleasant: it reduces utility by 1. If a bamboo-cutter does not get bitten, his income is  $Y_G$ , if he does get bitten by a snake, his income is  $Y_B$ .

- (a) What is the expected utility of a bamboo-cutter who takes precautions? And what is it if he does not take precautions?

Now suppose that  $Y_G = 4$  and  $Y_B = 0$ .

(b) Show that a bamboo-cutter optimally takes precautions.

Now suppose that bamboo-cutters can insure themselves against the loss of income after a snake-bite. An insured bamboo-cutter pays a premium  $p$ . If he gets bitten, the bamboo-cutter receives a payment of 4 from the insurance company.

(c) Show that an insured bamboo-cutter does not take precautions.

The insurance market is perfectly competitive, such that premiums are actuarially fair (i.e. the insurance companies make no profits, in expected terms). However, insurance companies cannot determine whether bamboo-cutters take precautions or not.

(d) Calculate the premium. Is it optimal for bamboo-cutters to buy insurance?

(e) Show that if bamboo-cutters would be able to commit to taking precautions, and the premium would adjust to account for the lower probability of snake-bites, then bamboo-cutters would be better off. Discuss.

Now suppose that there is a second group of bamboo-cutters. These people are identical to the first group, except that they are born with a special ability for avoiding snakes. This implies that their probability of being bitten is only 0.1 (regardless of precautions). The two groups are equally large.

(f) If insurance companies would be able to distinguish between the two groups, would the second group insure themselves against the income loss due to snake-bites?

(g) If insurance companies would *not* be able to distinguish between the two groups, would the second group insure themselves against the income loss due to snake-bites?

### Answers to exercises week 3

Rosen 9.5 / 9.6

- a)  $P = 100 - 25Q$ , with  $P = 50$  gives  $Q = 2$  visits per year. Total cost is 100.
- b) With 50 percent coinsurance, the individual pays \$25 per visit and the quantity demanded is 3 visits per year. The individual's out-of-pocket costs are \$75 and the insurance company pays \$75 (\$25 per visit, 3 visits per year).
- c) The introduction of insurance caused the quantity demanded to increase from 2 to 3 because the individual's effective price fell from \$50 to \$25, but the marginal cost is still \$50 per visit. The individual consumes medical services past the point where the marginal benefit to the individual equals the marginal cost, leading to inefficiency or deadweight loss. The deadweight loss (triangle) is equal to \$12.50.
- d) If the marginal external benefit of visiting the doctor is \$50, the socially optimal outcome is  $Q = 4$ . As the individual still chooses  $Q = 3$ , there is a suboptimal outcome now (i.e. the subsidy is below the marginal external benefit). Compared to the social optimum, this outcome is still \$12.50 less efficient.

Rosen 9.6 / 9.7

- a) There is a 95 percent chance of no illness, in which case income is \$30,000, and a 5 percent chance of illness, in which case income is \$10,000 because of the \$20,000 loss. Thus, expected income = 29,000. The utility of having a certain income of \$29,000 is 11.66, but the expected utility of this uncertain income is only  $0.95 \cdot U(30,000) + 0.05 \cdot U(10,000) = 11.64$ .
- b) An actuarially fair premium would be \$1,000 since there is a one in twenty chance that the insurance company will have to cover losses of \$20,000. If the individual buys insurance for \$1,000, then they have certain income of \$29,000 and the utility of \$29,000 is 11.66.
- c) Setting the expected utility equal to 11.64 and solving for income yields approximately \$28,388, indicating that the individual is indifferent between bearing the risk and having expected income of \$29,000 or purchasing insurance with a certain income of \$28,388. Hence, if the insurance costs  $\$30,000 - \$28,388 = \$1,612$ , the individual is indifferent between having insurance and not having insurance.

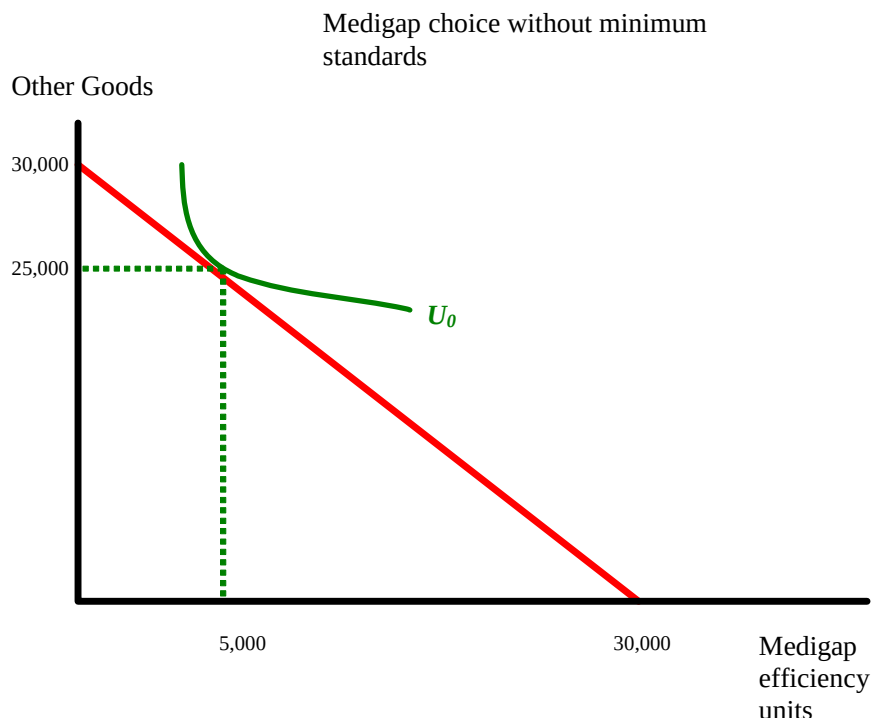
Rosen 9.7 / 9.8

In both options, the expected fine is 20. Denoting the value of littering by  $B$  and income by  $Y$ , the first option gives an expected value of littering equal to  $0.8U(B, Y) + 0.2U(B, Y - 100)$ , whereas the expected value of littering with the second option is  $0.9U(B, Y) + 0.1U(B, Y - 200)$ . Assuming diminishing marginal utility of income (i.e. risk-aversion with respect to income), expected utility is smaller in the second case than in the first case. Hence, the second option has a stronger deterrent effect,

leading to a larger reduction in littering. In addition, setting higher fines is cheaper than employing more police officers.

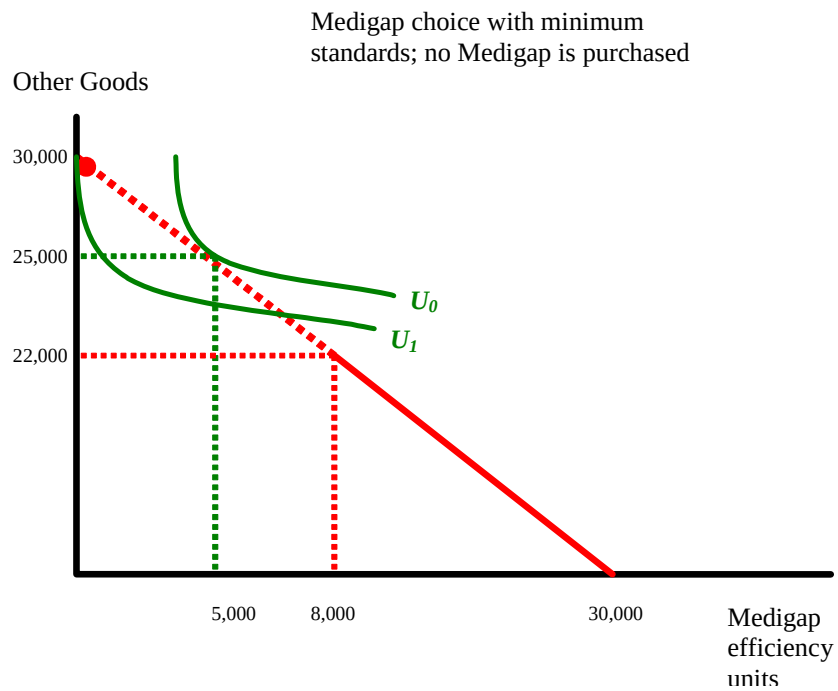
Rosen 10.4 / 10.5

The budget constraint initially has units of Medigap on the x-axis, and other goods on the y-axis. Given initial prices of \$1 per unit for each good, and \$30,000 of income, the budget constraint has a slope of -1, and the intercepts on both axes are at 30,000 units. It is assumed that the initial utility maximizing bundle consumes 5,000 units of Medigap, hence the indifference curve is tangent at (5000,25000). All of this is illustrated in the figure below.



After the “minimum Medigap” mandate, the consumer can either choose 0 units of Medigap or 8,000 or more units of Medigap. Thus, part of the budget constraint is eliminated (though the overall shape remains the same as before). After the mandate, the point (0,30000) is available, as well as all of the points to the southeast of the point (8000,22000). Clearly, the person’s utility must fall since the preferred choice, (5000,25000) is no longer available. If the person attains a higher level of utility as (0,30000) compared with (8000,22000), the person chooses to not purchase Medigap. In this case, the marginal rate of substitution is no longer equal to the price ratio.

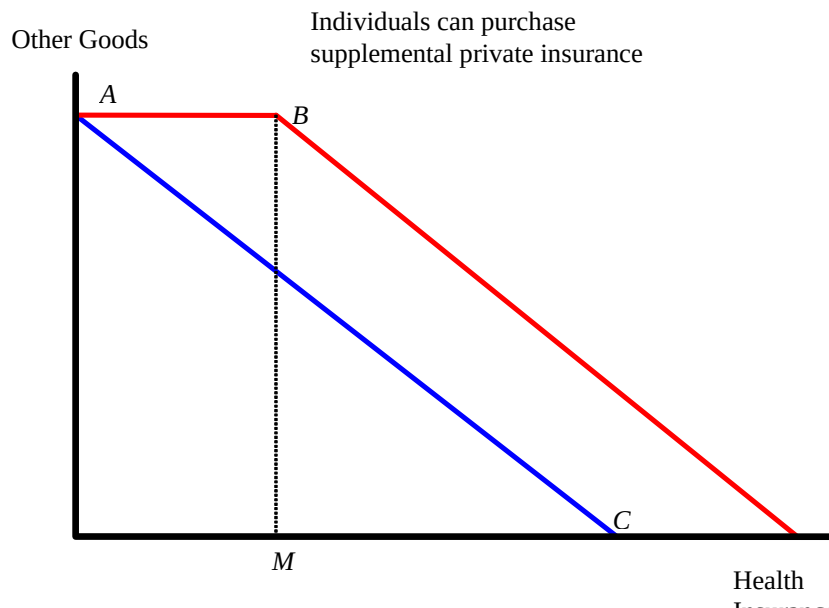
This is illustrated below.



Rosen 10.4(b)

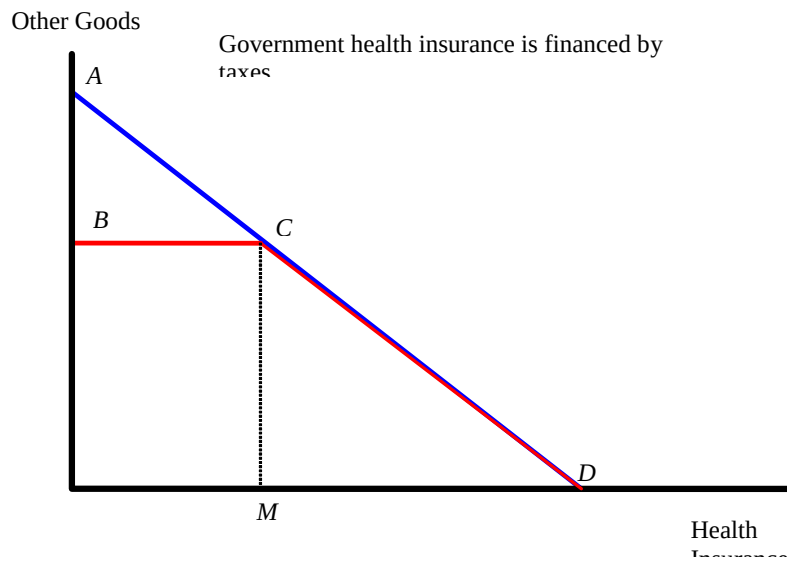
Rosen 10.5 / 10.6

If individuals are allowed to purchase supplemental private insurance, then the budget constraint in Figure 10.5 of the textbook is modified by drawing a line starting at point B that runs to the southeast and is parallel to AC.



Rosen 10.5(b)

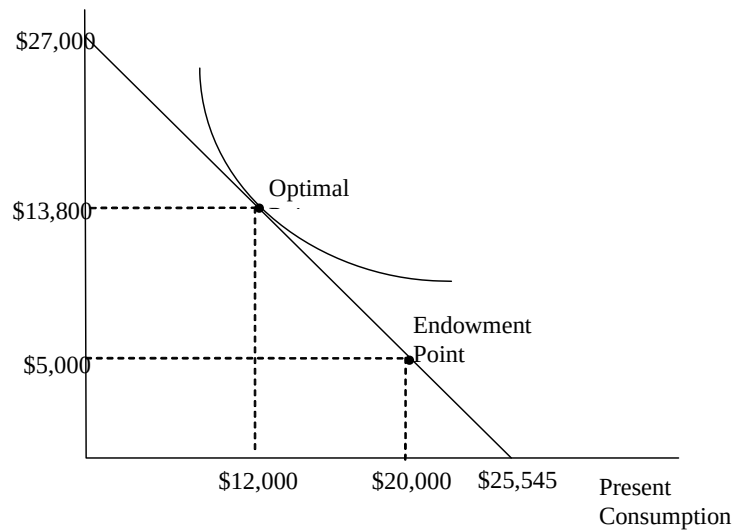
If individuals pay for health insurance (rather than perceiving it as being free), and the insurance was paid for with a lump sum tax, then the budget constraint shifts in by an amount that depends on the household's share of the tax burden. If the household's tax burden exactly equals the cost of health insurance, the budget constraint is no longer the line segment AD but rather the segment BCD, where the segment CD runs along the original budget constraint, except that the minimum amount of health insurance consumed is M.



Rosen 10.5(b)

Rosen 11.8 / 11.8

a) The problem does not provide information about the utility function, so the optimal point is where the indifference curve is drawn tangent to the budget line, which can occur at different values depending on how the curve is drawn. In the diagram below, the optimal point involves saving \$8,000 and future consumption consists of period 2 income (\$5,000) plus savings with interest (\$8,800). (note that the present value of consumption is \$24,525, rather than the \$25,545 specified in the figure



Rosen 11.8

- b) If Social Security takes \$3,000 from the individual in the first period and pays him this amount with interest in the second period, then private savings falls from \$8,000 to \$5,000. There would be no change in optimal consumption values.

### Exercise 3.1

Drawing the uniform distribution helps!

a) Average revenue equals  $\frac{1}{2}([5000 \cdot 0 + 1000] + [5000 \cdot 1 + 1000]) = 3500$  (uniform distribution of ability). Hence, when all employees work for a company, on average they create 3500 in revenue, so that the company is maximally willing to pay a wage equal to 3500.

b) Given wage  $w = 3500$ , all individuals with ability  $a_i$  such that  $5000 \cdot a_i \geq 3500$  prefer to be self-employed. Rewriting yields  $a_i \geq 0.7$ . Hence, the most productive employees leave the company: adverse selection.

c) Companies anticipate upon the above: when they would pay a wage equal to 3500, the average revenue of employees would only be  $\frac{1}{2}([5000 \cdot 0 + 1000] + [5000 \cdot 0.7 + 1000]) = 2750$ , implying that companies would make a loss (on average). Hence, companies pay lower wages, which in turn implies that more (relatively able) individuals choose to be self-employed.

To find the equilibrium, denote the ability of individuals who are just indifferent between being self-employed and being employee by  $a^*$ . Hence,  $a^*$  is given by  $w^* = 5000 \cdot a^*$ , where  $w^*$  is companies' wage in equilibrium. The wage must be such that the average productivity of all employees is equal to this wage (zero-profit). This

gives:  $w^* = \frac{1}{2}([5000 \cdot 0 + 1000] + [5000 \cdot a^* + 1000])$ . Substituting for  $w^* = 5000 \cdot a^*$  gives

$$5000 \cdot a^* = \frac{1}{2}([5000 \cdot 0 + 1000] + [5000 \cdot a^* + 1000])$$

which can be rewritten to

$$\frac{1}{2}5000 \cdot a^* = \frac{1}{2}(1000 + 1000) \rightarrow a^* = \frac{1000}{2500} = 0.4$$

Which implies that equilibrium wage  $w^* = 2000$ . Hence, the least able 40% of all individuals works for companies and earns a wage of 2000, the rest is self-employed.

d) Efficiency requires that all individuals work for a company (revenue created is 1000 higher). Inefficiency arises because of private information. If information can be (credibly) shared, outcome may be improved. Two ways to achieve this are:

- Pay-for-performance (or screening): if companies can make individuals' wages depended on the revenue they create, the adverse selection problem does not arise. See Personnel Economics for more info.

- Signalling: If individuals can somehow credibly signal their ability (for instance by doing studies which are more easy to finish with high ability), companies can pay higher wages to high-ability individuals.

Note: both of these possible solutions (can) have negative effects: pay-for-performance may not work perfectly (risk, distortions), which reduces efficiency, and when signalling is costly, there may be too much signalling (from an efficiency perspective).

### Exercise 3.2

a) When price  $P \geq 3000$ , all cars will be traded. Then, expected value for buyer is 4000. Hence, maximum price they are willing to pay is 4000. When the price  $P$  would be lower than 3000, high-quality cars would not be sold, and buyers would anticipate this. Hence, all cars are sold, for a price between 3000 and 4000.

b) Yes, for both good and bad cars, buyers' valuation is higher than sellers' valuation.

c) If both types are offered, then the expected value for a buyer is 2800. Hence, if all cars would be traded, buyers would be willing to pay maximally 2800. At that price, sellers of good cars do not offer their car for sale. Hence, only low quality cars are sold. Buyer anticipate this, which implies that they will never pay more than 1600. Hence, the price lies somewhere between 1000 and 1600.

d) Buyers can determine the quality of the car, at the cost of 336. Hence, including the inspection cost, their valuation of a bad car is 1264 and of a good car 3664. Now, (before inspection) buyer and seller can agree to a price between 1000 and 1264 if the car turns out to be bad and a price between 3000 and 3664 if the car turns out to be good. Even though all cars are being traded now, which is efficient,

the inspection cost are a waste. Hence, the inspection cost are a necessary bad in order to avoid an even greater inefficiency: the withdrawal of good quality cars from the market.

e) It is better to have the cars inspected only when the seller claims that the car is good. Since sellers know the quality of their car, inspecting bad cars is not necessary. Hence, a better solution is that sellers of good cars have their cars inspected, and that inspected cars trade for a price between 3336 and 4000. Bad cars are not inspected, and are traded for a price between 1000 and 1600.

f) As inspection is costly, less inspections is better (i.e. more efficient). However, if no car is inspected and the price of a good car would be higher than the price of a bad car, all sellers would claim that their car is good. Here, we look for the minimum probability of inspection of cars claimed to be good by their sellers, denoted by  $x$ , at which sellers of bad cars still do not have an incentive to claim that their car is good. Suppose the price of a good car is  $p_G$  and the price of a bad car is  $p_B$ . If a seller claims the car is good, the probability that the car is inspected is  $x$ . Cars are sold as good cars if they are inspected and found good, as well as when the seller claims the car is good but the car is not inspected. Clearly, a seller of a good car will always claim that the car is good. Consider a seller of a bad car. If he claims the car is bad, he gets price  $p_B$ . If he claims the car is good, either the car is not inspected (with probability  $1 - x$ ), in which case he gets  $p_G$ , or the car is inspected (which happens with probability  $x$ ), yielding price  $p_B - 336$  (inspection cost). Hence, the seller of a bad car is honest when

$$p_B \geq (1 - x)p_G + x[p_B - 336] \rightarrow (1 - x)p_B \geq (1 - x)p_G - 336x$$

To find the minimum  $x$  for which this inequality holds (so that inspections are minimised), we should substitute for the highest possible  $p_B$  and the lowest possible  $p_G$ . Buyers maximally pay 1600 for a bad car, sellers are willing to sell a good car at minimally  $3000 + 336x$  (as in expectation, they have to pay  $336x$  in inspection cost). This gives

$$\begin{aligned} (1 - x)1600 &\geq (1 - x)[3000 + 336x] - 336x \\ 0 &\geq 1400(1 - x) - 336x^2 \end{aligned}$$

which can be solved for  $x \geq \frac{5}{6}$  (using a good calculator/math program).

Thus, in order to maximise efficiency in this market, it should be that  $p_B = 1600$ ,  $p_G = 3000 + 336 \cdot \frac{5}{6} = 3280$ , and the probability of inspection in case the seller claims the car is good  $= \frac{5}{6}$ .

g) Part of the higher price of the dealer is the inspection cost and the warranty. The information dealers provide about quality is valuable.

### Exercise 3.3

a) Expected utility with precautions:  $EU_P = 6\sqrt{Y_G} + 2\sqrt{Y_B} - 1$ .

Expected utility without precautions:  $EU_{NP} = 4\sqrt{Y_G} + 4\sqrt{Y_B}$ .

b) Expected utility with precautions:  $EU_P = 6\sqrt{4} + 2\sqrt{0} - 1 = 11$ .

Expected utility without precautions:  $EU_{NP} = 4\sqrt{4} + 4\sqrt{0} = 8$ .

c) Expected utility with insurance and precautions:  $EU_P^I = 6\sqrt{4-p} + 2\sqrt{4-p} - 1 = 8\sqrt{4-p} - 1$ .

Expected utility with insurance but without precautions:  $EU_{NP}^I = 4\sqrt{4-p} + 4\sqrt{4-p} = 8\sqrt{4-p}$ .

d) The premium  $p$  should be equal to the expected payout, given that bamboo-cutters do not take precautions:  $p = \frac{1}{2}0 + \frac{1}{2}4 = 2$ . Hence, expected utility with insurance (and without precautions) is  $EU_{NP}^I = 8\sqrt{2} = 11.3$ . Expected utility without insurance is  $EU_P^{NI} = 11$  (note that under (b), we derived that without insurance, they would take precautions). Hence, they prefer to buy insurance.

e) If bamboo-cutters could (credibly) commit to taking precautions, the premium would be  $p = \frac{3}{4}0 + \frac{1}{4}4 = 1$ . Hence,  $EU_P^I = 8\sqrt{3} - 1 = 12.9$ . Thus, although moral hazard makes bamboo-cutters better off given that they have insurance, the fact that the insurance companies anticipate this moral hazard and adjust their premium implies that bamboo-cutters are actually worse off due to their own moral hazard behaviour. They would prefer to be able to commit themselves to always taking precautions.

f) Premium for the second group would be  $p_2 = 0.9 \cdot 0 + 0.1 \cdot 4 = 0.4$ . Expected utility without insurance is  $EU = 0.9 \cdot 8\sqrt{4} + 0.1 \cdot 8\sqrt{0} = 14.4$ . Expected utility with insurance equals  $EU = 0.9 \cdot 8\sqrt{4-0.4} + 0.1 \cdot 8\sqrt{4-0.4} = 15.2$ .

g) There can be only one premium. If everyone is insured, expected payout is given by  $\frac{1}{2} [\frac{1}{2}0 + \frac{1}{2}4] + \frac{1}{2} [0.9 \cdot 0 + 0.1 \cdot 4] = 1.2$  (note: first part is expected payout to a member of group 1, second part expected payout to member of group 2). Expected utility of a member of group two with this insurance is  $EU = 0.9 \cdot 8\sqrt{4-1.2} + 0.1 \cdot 8\sqrt{4-1.2} = 13.4$ . This is smaller than their utility without insurance. Hence, group two will not insure themselves, implying that only group 1 insures, at a premium  $p = 2$ .

Note that this is not the efficient outcome, as groups 2 does benefit from insurance. This is an example of how asymmetric (or private) information can lead to adverse selection, and, thereby, to reduced efficiency.